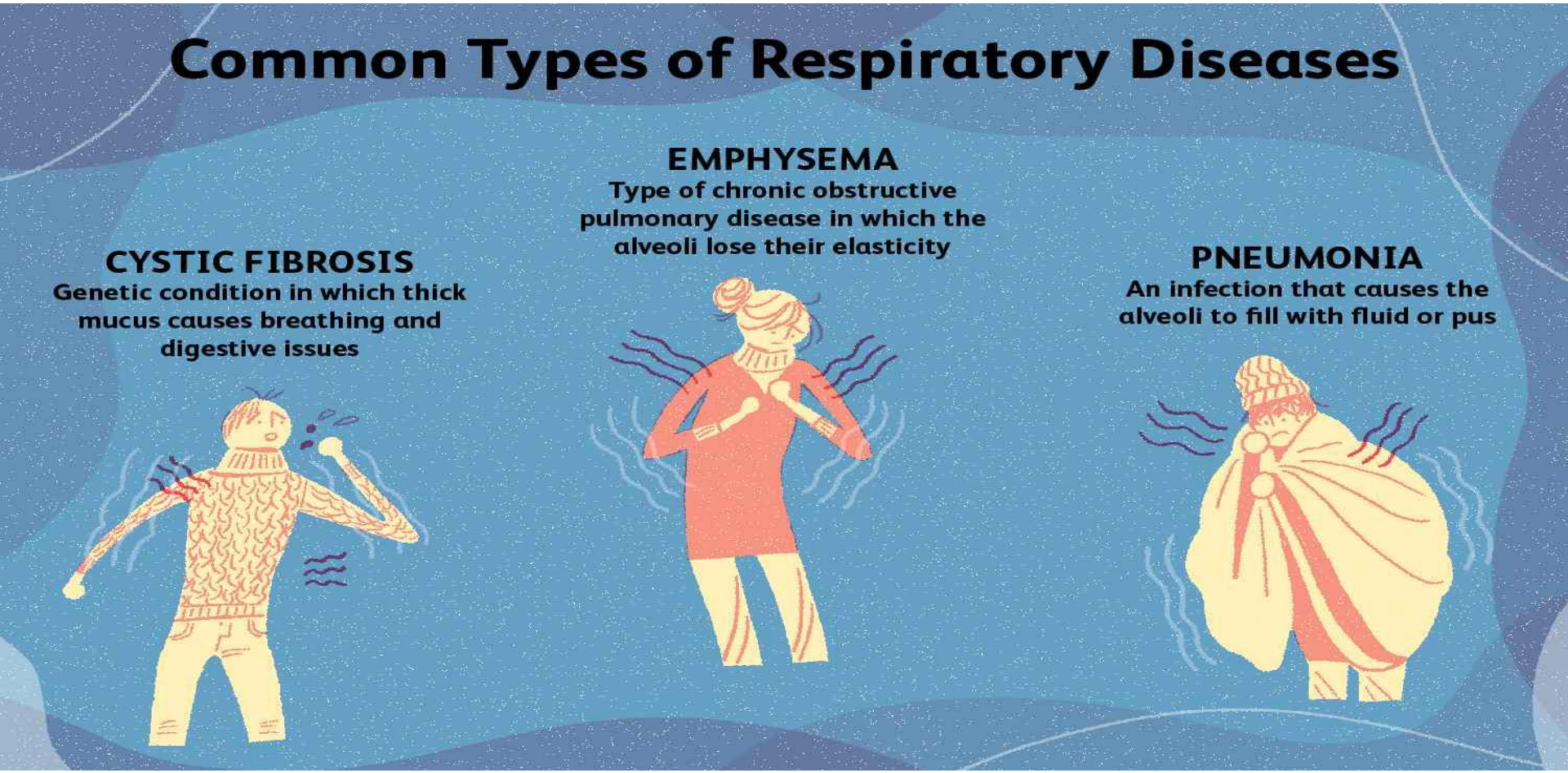
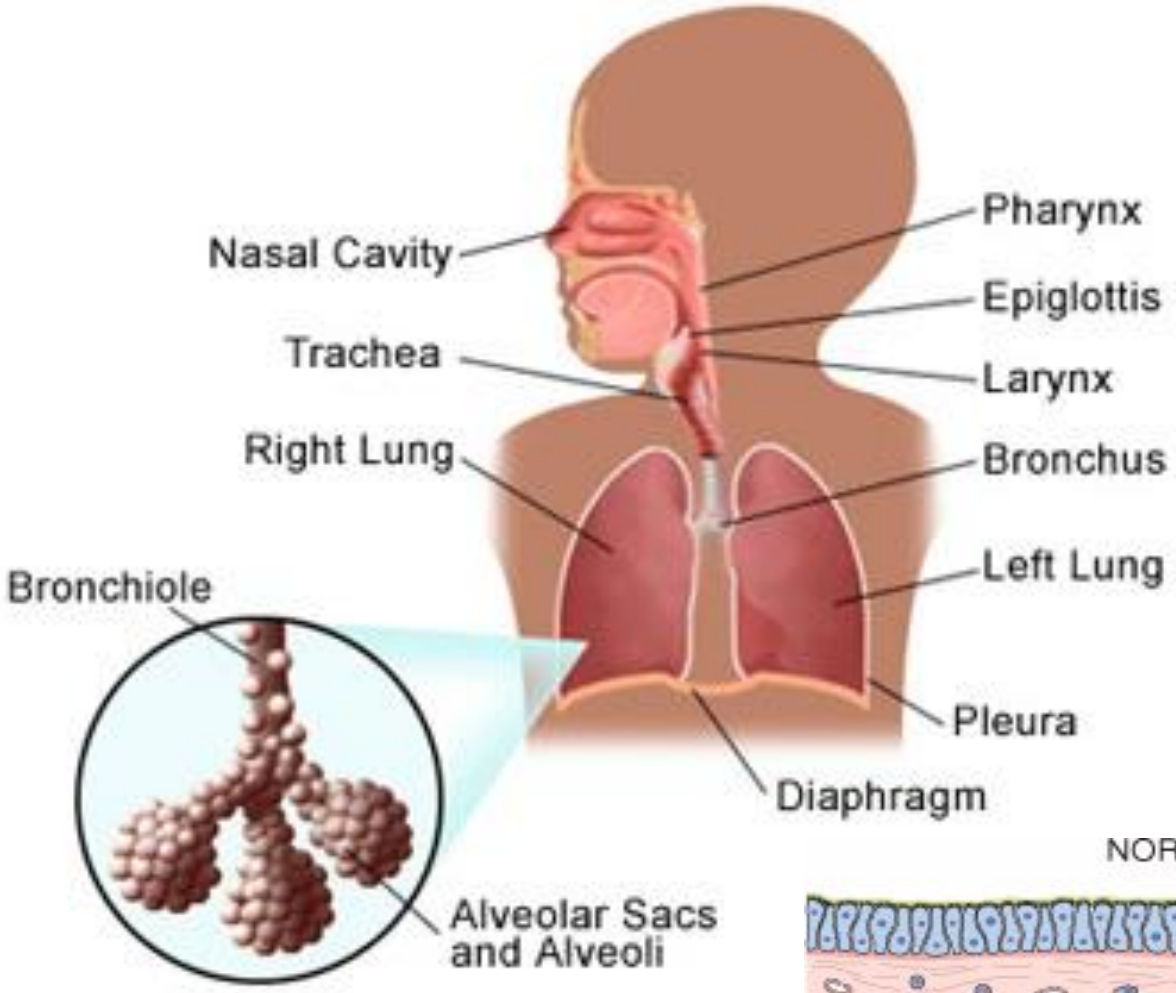
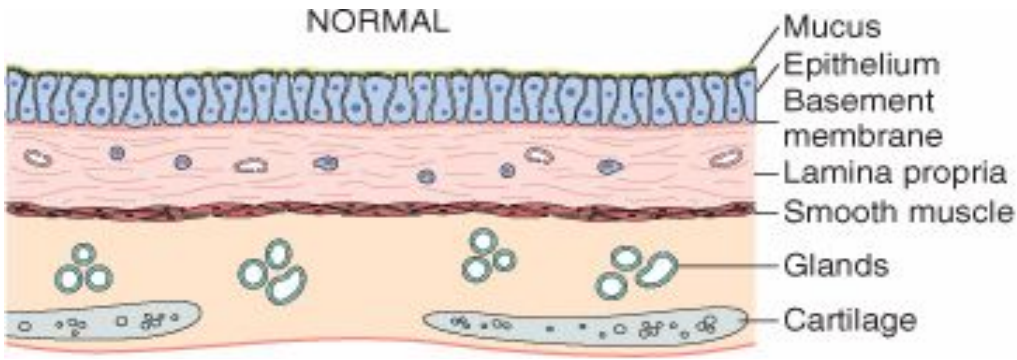
The Diseases of the Respiratory System



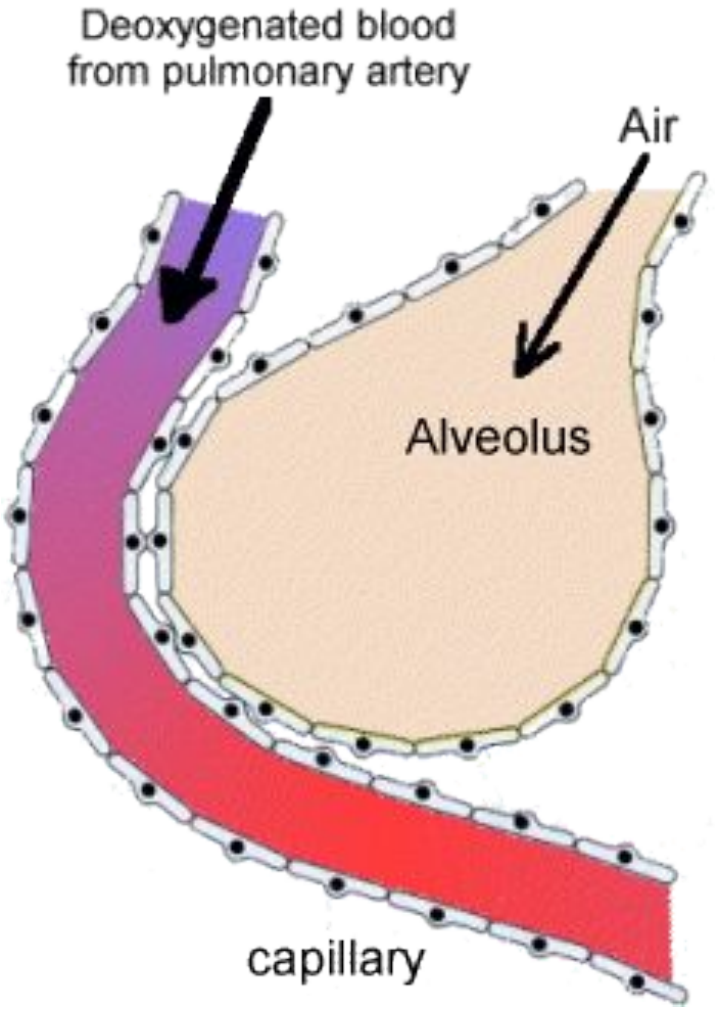
DR ABDUL SAMAD KHAN
SENIOR LECTURER
DIKIOHS-DUHS

Introduction: Anatomy





Introduction: Physiology



IMPORTANT TERMS USED FOR SOME OF THE ALTERED PATTERNS OF RESPIRATION

- **EUPNEA** : Refers to normal respiration
- **TACHYPNEA** : The increase in rate of respiration
- **BRADYPNEA** : The decrease in rate of respiration
- **APNEA :** Temporary arrest of breathing
- **HYPERPNEA** : Increase in pulmonary ventilation due to increase in rate or force of respiration more commonly occurs after exercise

HYPERVENTILATION : Abnormal increase in rate & force of respiration very often lead to dizziness, discomfort and chest pain

- **DYSPNEA :** Difficulty in breathing
- **HYPOXIA :** Defined as reduced availability of O_2 to the tissue
- **HYPERCAPNEA :** Increased CO_2 content of blood
- **CYANOSIS** : Is defined as diffused bluish coloration of skin and mucus membrane. It is due to the presence of large amount of reduced hemoglobin in the blood

Common symptoms

- **Dyspnea:** difficulty with breathing

Cause may be;

- Decrease compliance, fibrosis
- Increased airway resistance , chronic bronchitis
- Chest wall disease, obesity
- Fluid accumulation, left sided heart failure

- **Cough**

Cause may be:

- Postnasal discharge,
- GERD (**Gastroesophageal reflux disease**),
- Br. Asthma,
- chronic Bronchitis,
- pneumonia,
- bronchiectasis,
- drug induced

- **Hemoptysis**

Cause may be;

- TB, bronchiectasis,
- Ch. Bronchitis, pneumonia
- aspergilloma (Aspergillosis is an infection caused by the fungus aspergillus).

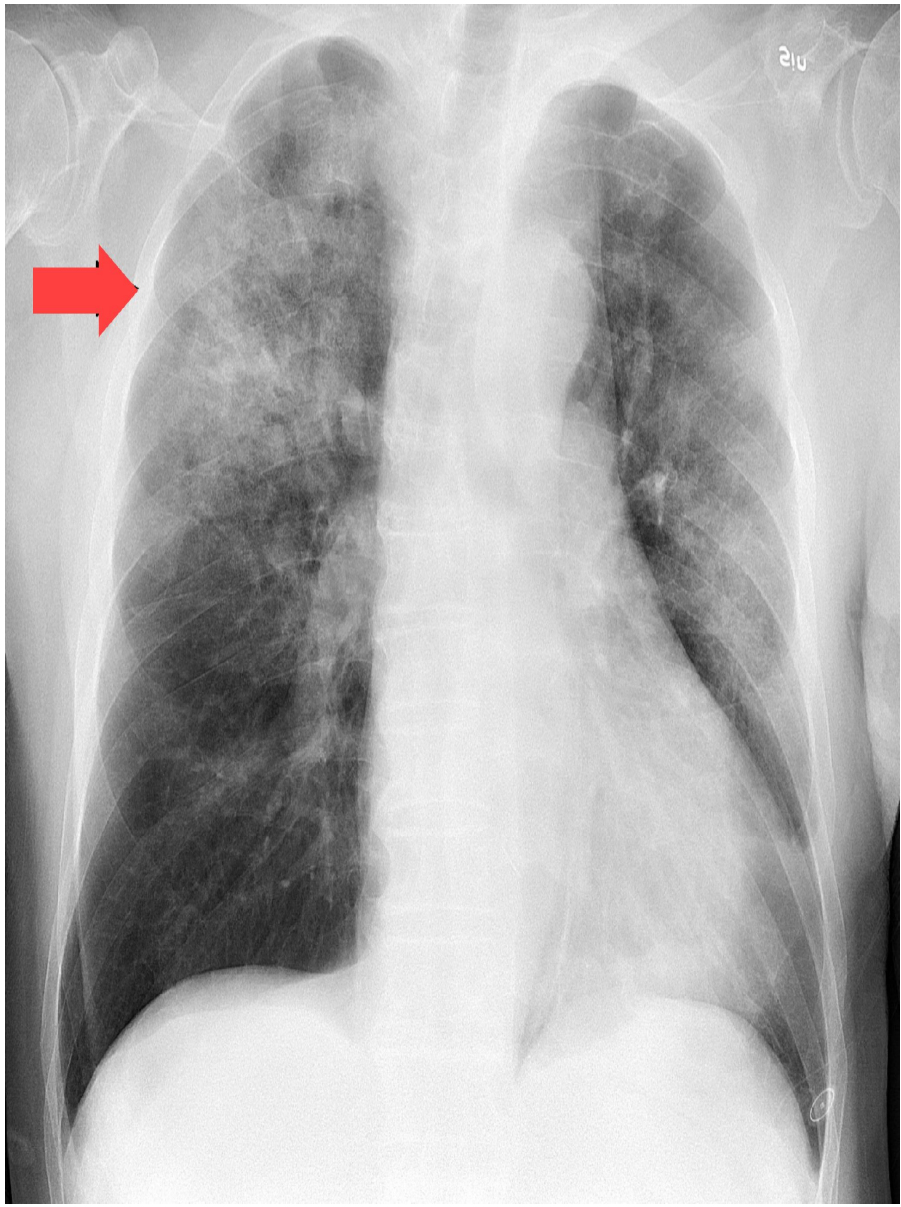
COMMON RESPIRATORY DISORDERS

PNEUMONIA

- Is the inflammation of lung tissues followed by the accumulation of blood cells, fibrin and exudates in the alveoli

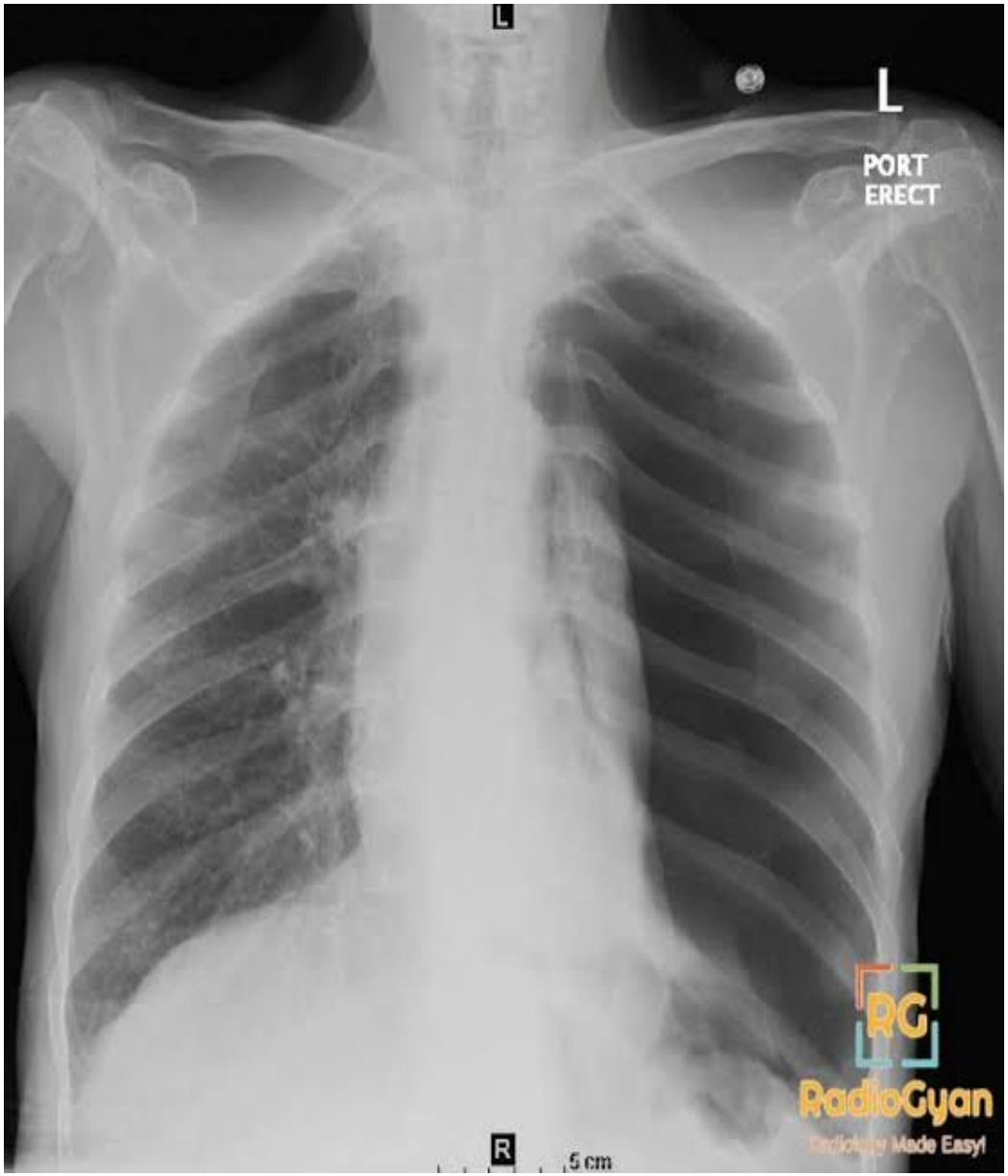
CAUSES:

- Bacterial or viral infection
- Inhaling noxious chemical substance



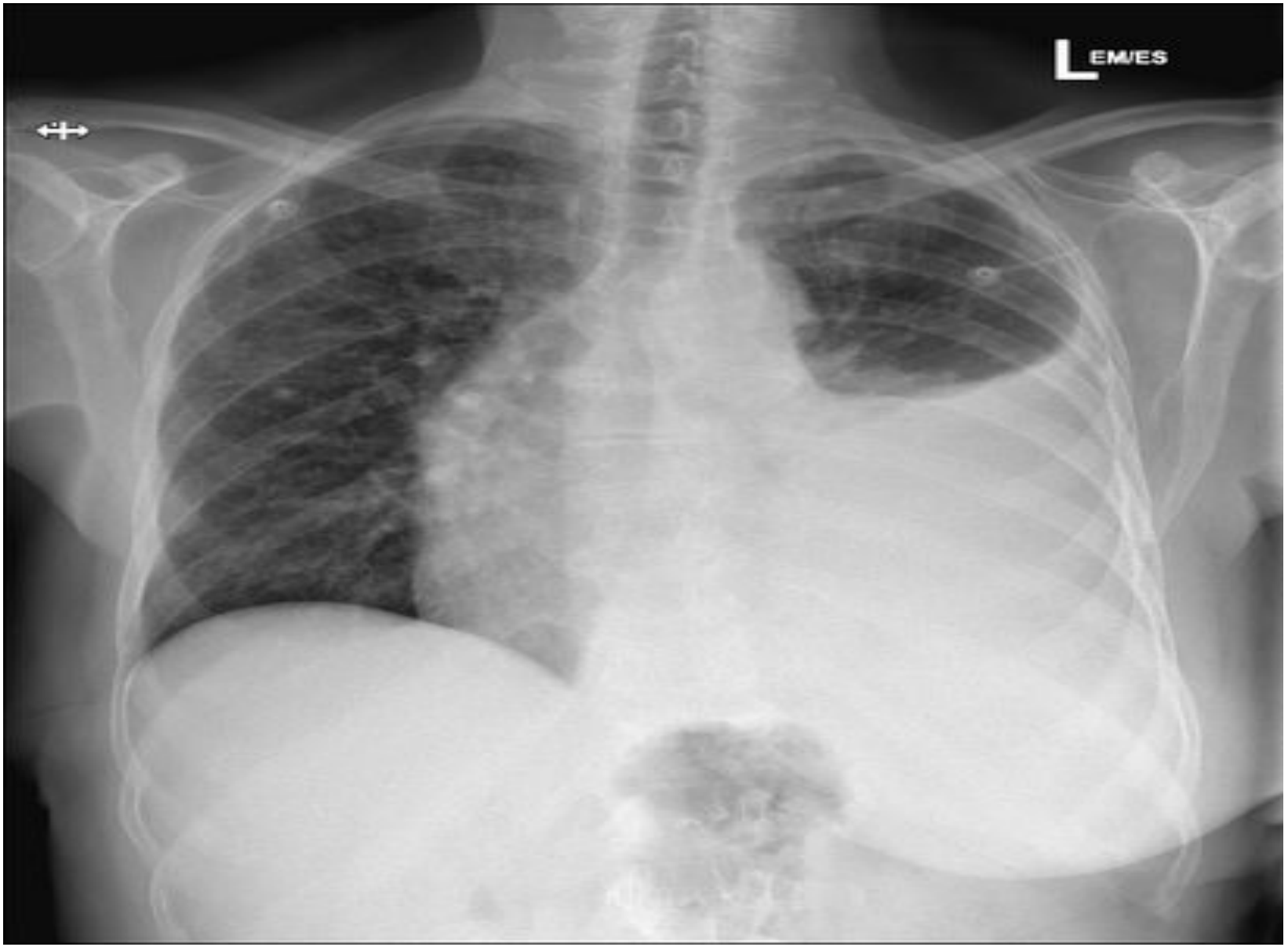
PNEUMOTHORAX

Presence of air in pleural space.



PLEURAL EFFUSION

Accumulation of large amount of fluid in the pleural cavity.



PULMONARY TUBERCULOSIS :

Tuberculosis is the disease caused by tubercle bacilli.

This disease can affect any organ in the body. However, the lungs are affected more commonly.

The infected tissue is invaded by macrophages and later it becomes fibrous.



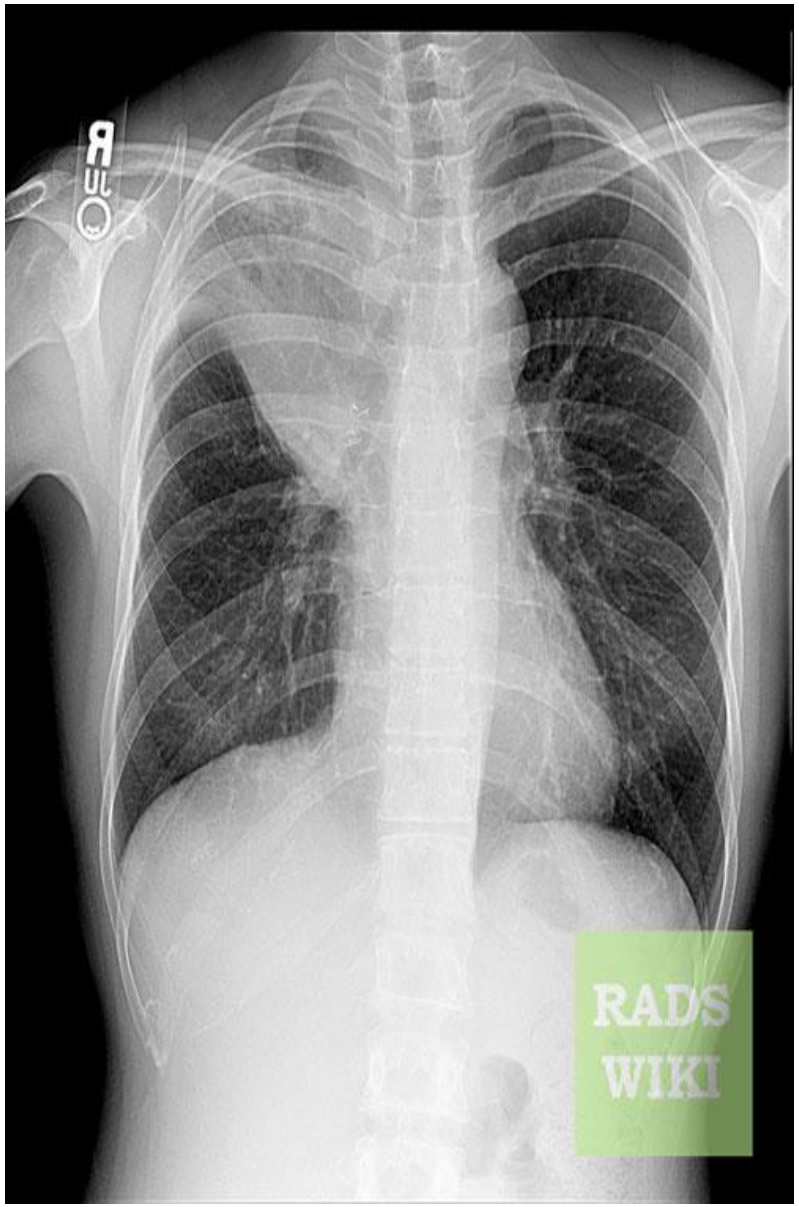
Atelectasis (collapse)

Incomplete expansion of the lungs or collapse of previously inflated lung substance.

Significant atelectasis reduce oxygenation and predispose to infection.

Types of Atelectasis

- Resorption atelectasis.
- Compression atelectasis.
- Contraction atelectasis



Types of Atelectasis

1- Resorption atelectasis

Result from complete obstruction of an airway and absorption of entrapped air.

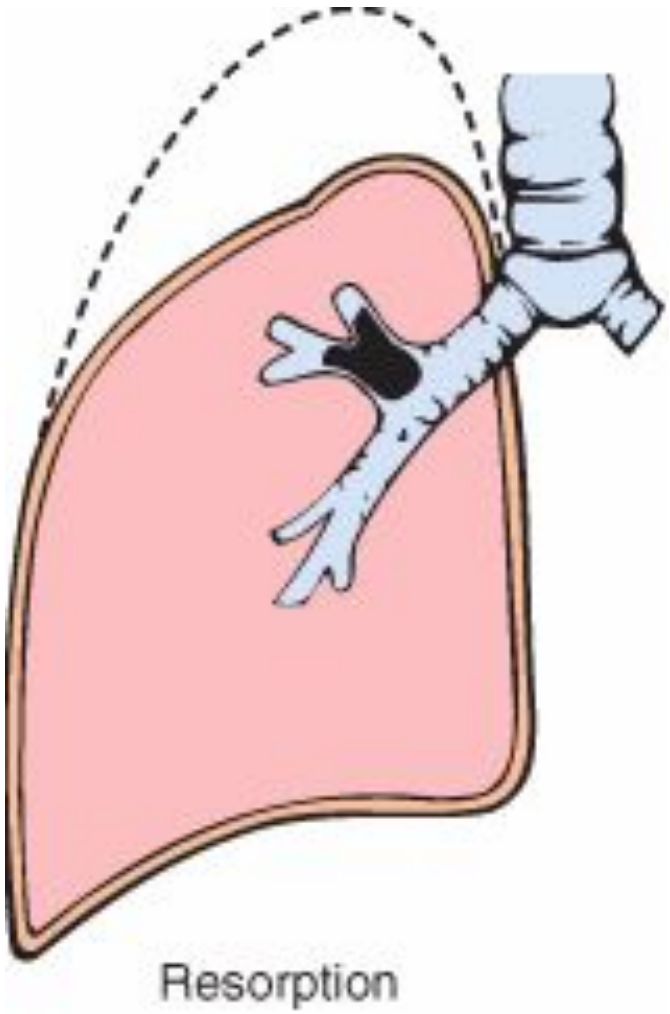
Obstruction can be caused by:

a. **Mucous plug** (postoperatively or exudates within small bronchi seen in bronchial asthma and chronic bronchitis).

b. Aspiration of foreign body

c. Neoplasm

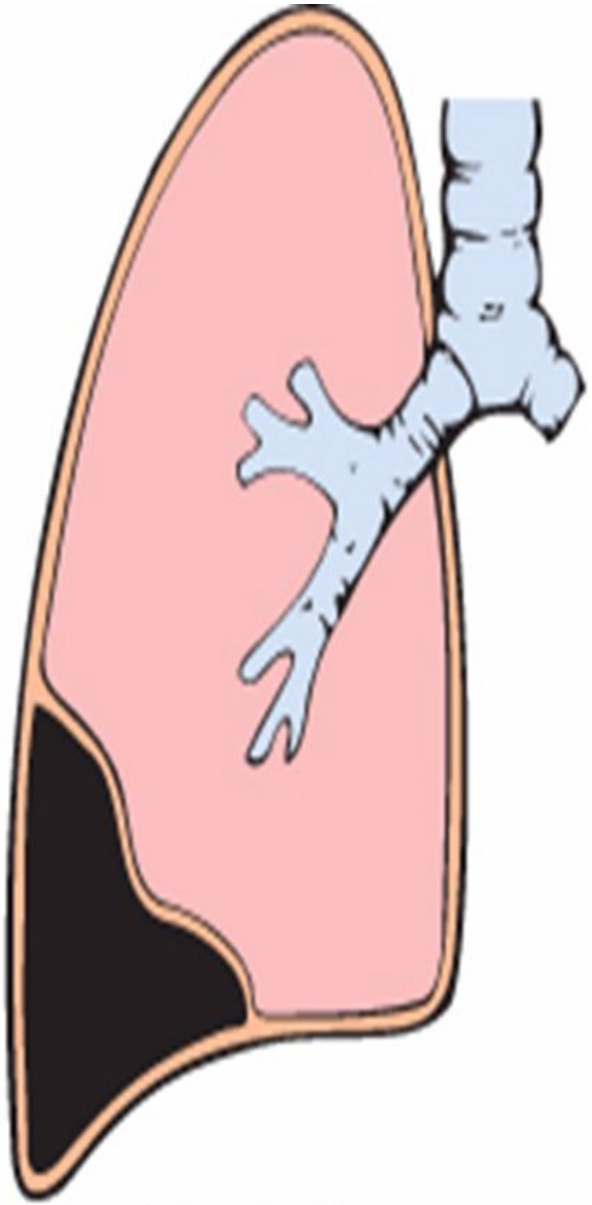
d. enlarged lymph node



2- Compression atelectasis

Results whenever the pleural cavity is partially or completely filled by fluid, blood, tumor or air, e.g.

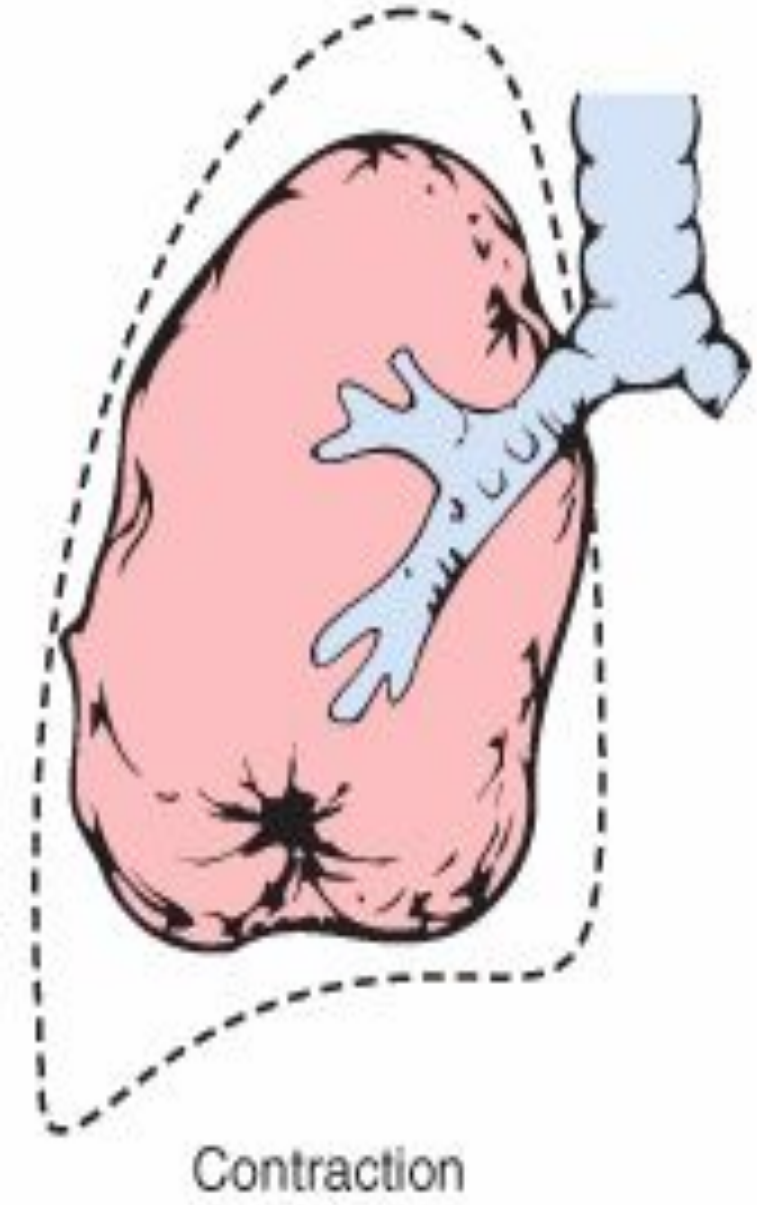
- patient with cardiac failure
- patient with neoplastic effusion
- patient with abnormal elevation of diaphragm in peritonitis or subdiaphragmatic abscess



Compression

3- Contraction atelectasis

Local or generalized fibrotic changes in pleura or lung preventing full expansion of the lung.



Atelectasis

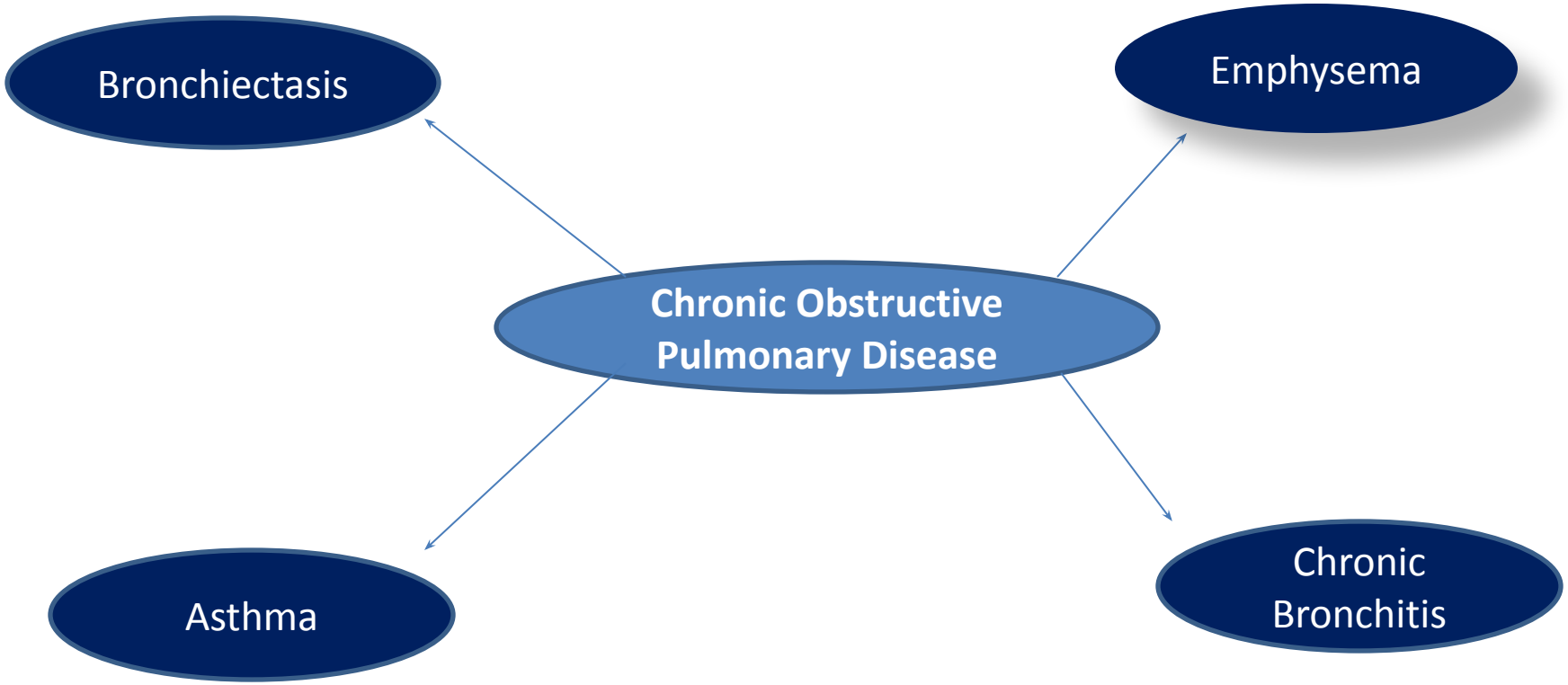
- Atelectatic lung is prone to develop superimposed infection.
- It is reversible disorder except for contraction atelectasis.
- It should be treated promptly to prevent hypoxemia.

Chronic Obstructive Pulmonary Disease (COPD)

Share a major symptom:

- Dyspnea with chronic or recurrent obstruction to airflow within the lung.
- The incidence of COPD has increased dramatically in the past few decades.

Chronic Obstructive Pulmonary Disease



A group of conditions characterized by limitation of airflow

Emphysema and chronic bronchitis often co-exist.

Emphysema

Emphysema: It is characterized by permanent enlargement of the airspaces distal to the terminal bronchioles accompanied by destruction of their walls, without obvious fibrosis

Types

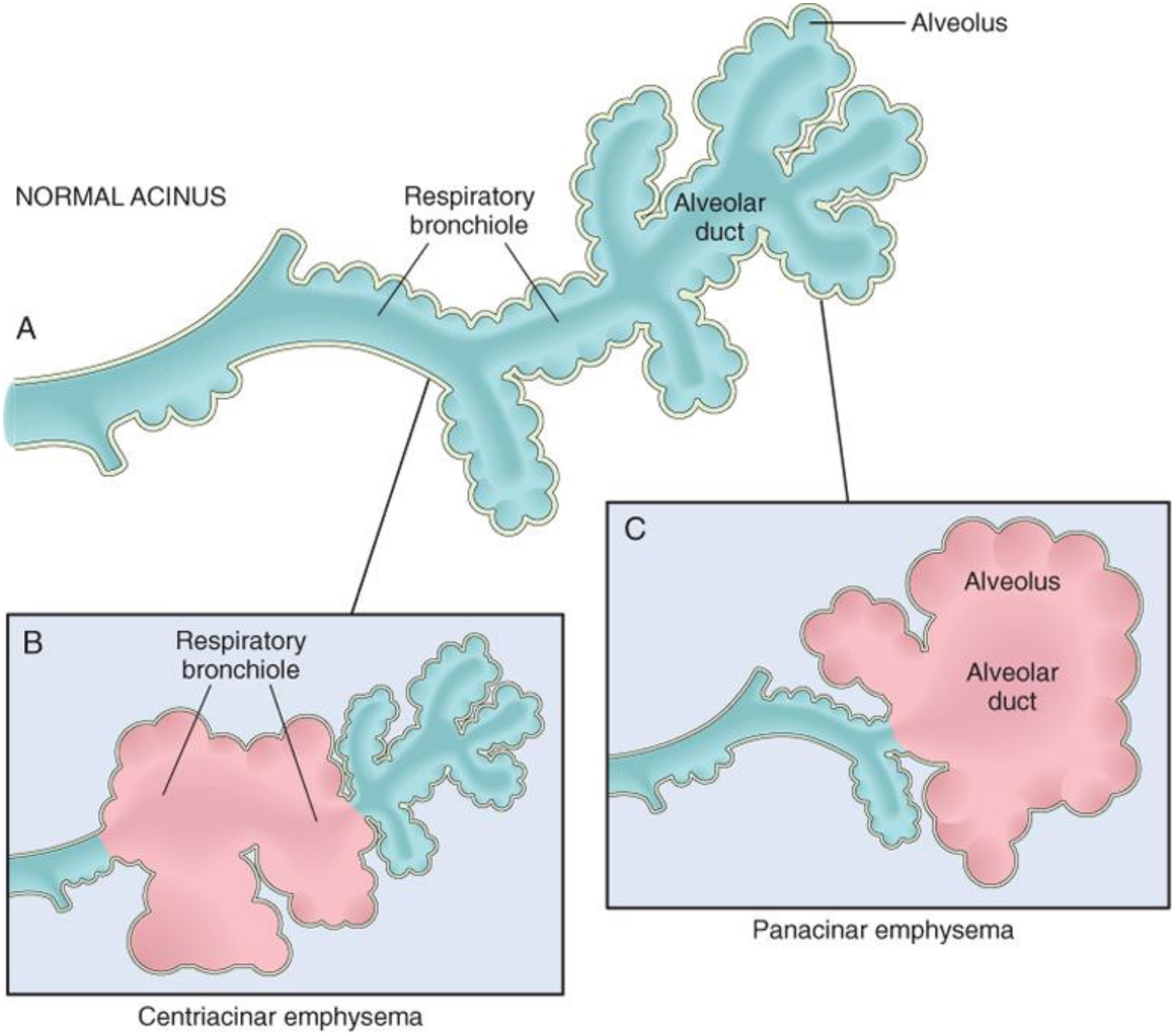
- Centriacinar: Smoking
- Panacinar: deficiency of α1 AT
- Paraseptal
- Irregular: scar

Clinical features

- Cough and wheezing.
- Respiratory acidosis
- Weight loss.
- Pulmonary function tests reveal reduced FEV1.

Complications

- Pneumothorax
- Death from emphysema is related to:
  - Pulmonary failure with respiratory acidosis, hypoxia and coma.
  - Right-sided heart failure (Cor pulmnale)



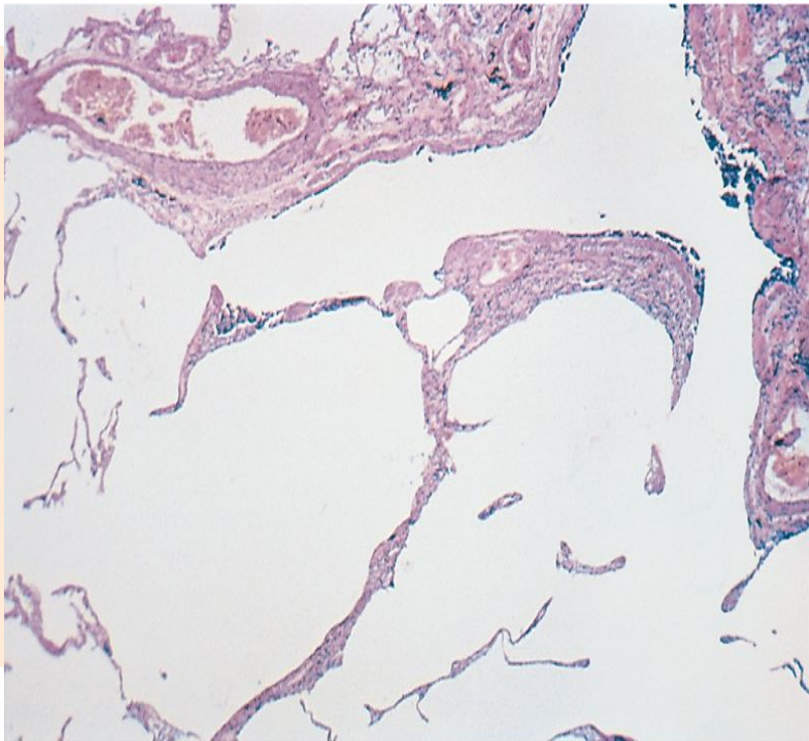
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Emphysema

Morphology

- The diagnosis depend largely on the macroscopic appearance of the lung.
- The lungs are pale, voluminous.
- Histologically, thinning and destruction of alveolar walls creating large airspaces.

- Loss of elastic tissue.
- Reduced radial traction on the small airways.
- Alveolar capillaries is diminished.
- Fibrosis of respiratory bronchioles.
- Accompanying bronchitis and bronchiolitis.



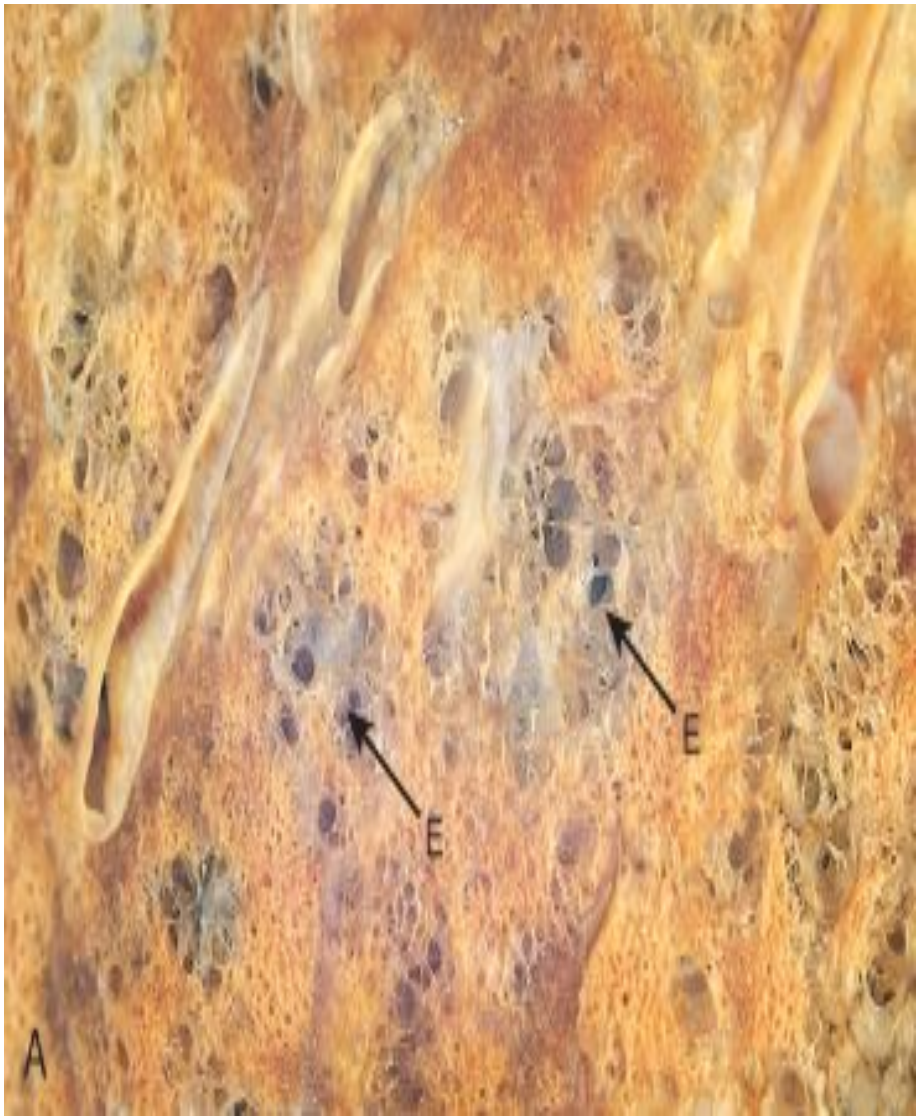
Emphysema

Incidence

- Emphysema is present in approximately 50% of adults who come to autopsy.
- The prevalence of emphysema in the United States is approximately 14 million, which includes 14% white male smokers and 3% white male nonsmokers.
- Pulmonary disease was considered to be responsible for death in 6.5% of these patients.

Centriacinar (centrilobular) emphysema

- Occur in heavy smoker in association with chronic bronchitis
- The central or proximal parts of the acini are affected, while distal alveoli are spared



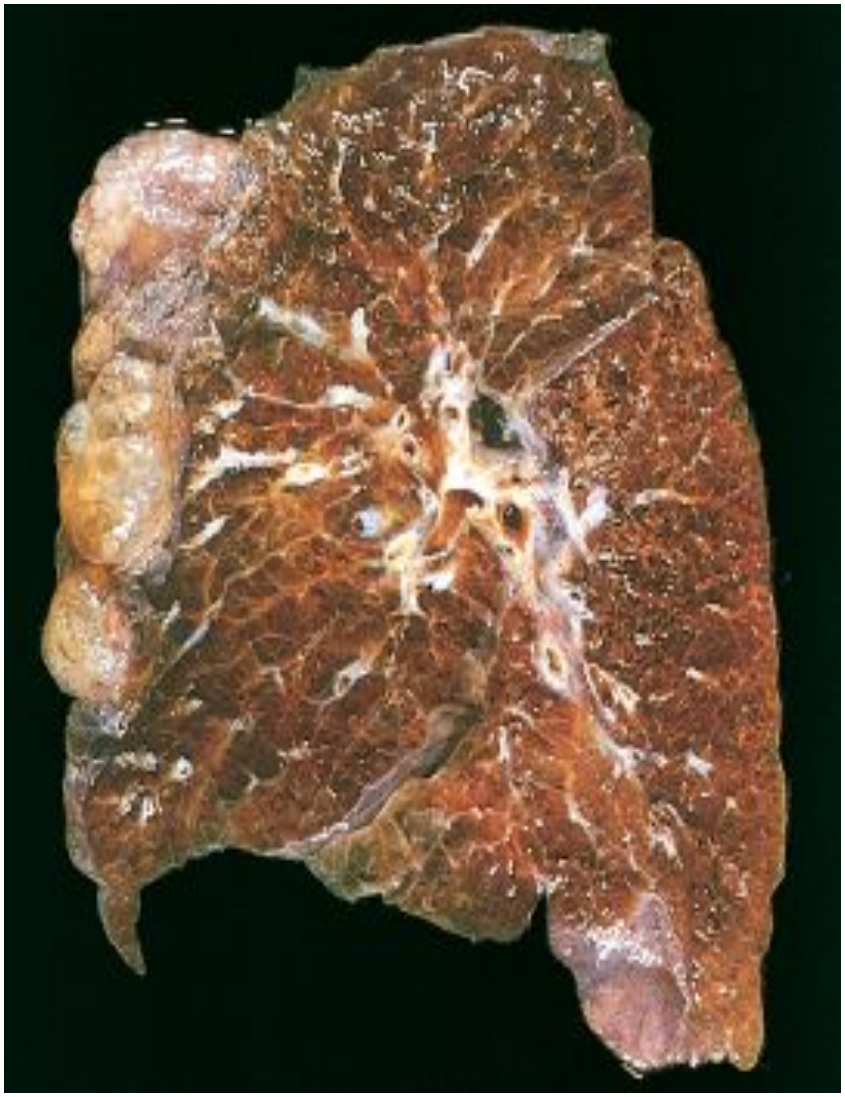
Panacinar (panlobular) emphysema

- Occurs in α_1-anti-trypsin deficiency. (AAT is a protein made in your liver to help protect the lungs.)
- Acini are uniformly enlarged from the level of the respiratory bronchiole to the terminal blind alveoli.
- More commonly in the lower lung zones.



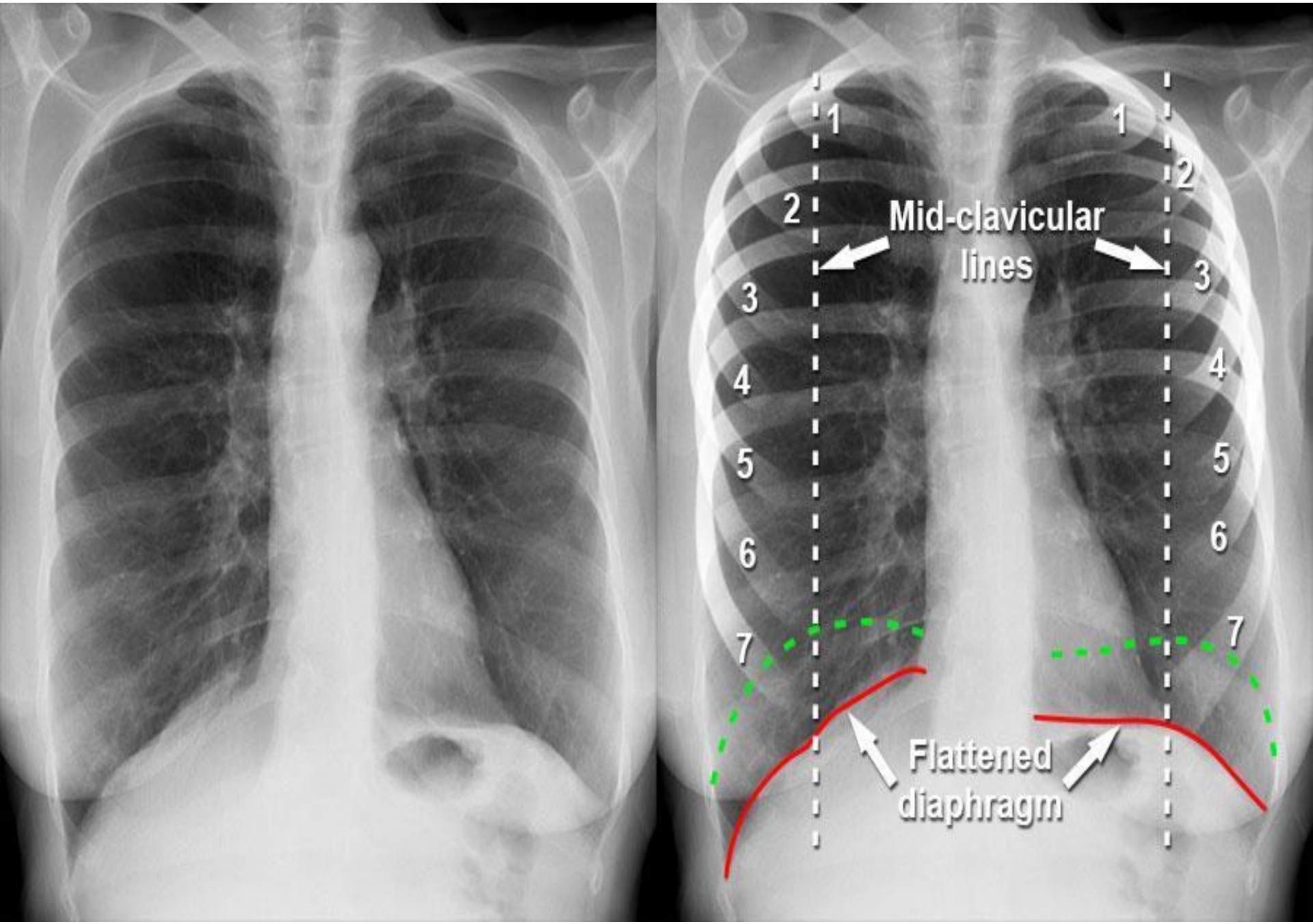
Distal acinar (paraseptal) emphysema

- The proximal portion of the acinus is normal but the distal part is dominantly involved.
- Occurs adjacent to areas of fibrosis, scarring or atelectasis.
- More severe in the upper half of the lungs.
- Sometimes forming multiple cyst-like structures with spontaneous pneumothorax.



Irregular Emphysema

- The acinus is irregularly involved, associated with scarring.
- Most common form found in autopsy.
- Asymptomatic.


1
2
3
4
5
6
7
Mid-clavicular lines
Flattened diaphragm

Chronic Bronchitis

Chronic Bronchitis

- Persistent productive cough for at least 3 consecutive months in at least 2 consecutive years, smoking related
- Common among cigarette smokers and urban dwellers, age 40 to 65

Types:

1. Simple chronic bronchitis.
2. Chronic mucopurulent bronchitis.
3. Chronic asthmatic bronchitis.
4. Chronic obstructive bronchitis.

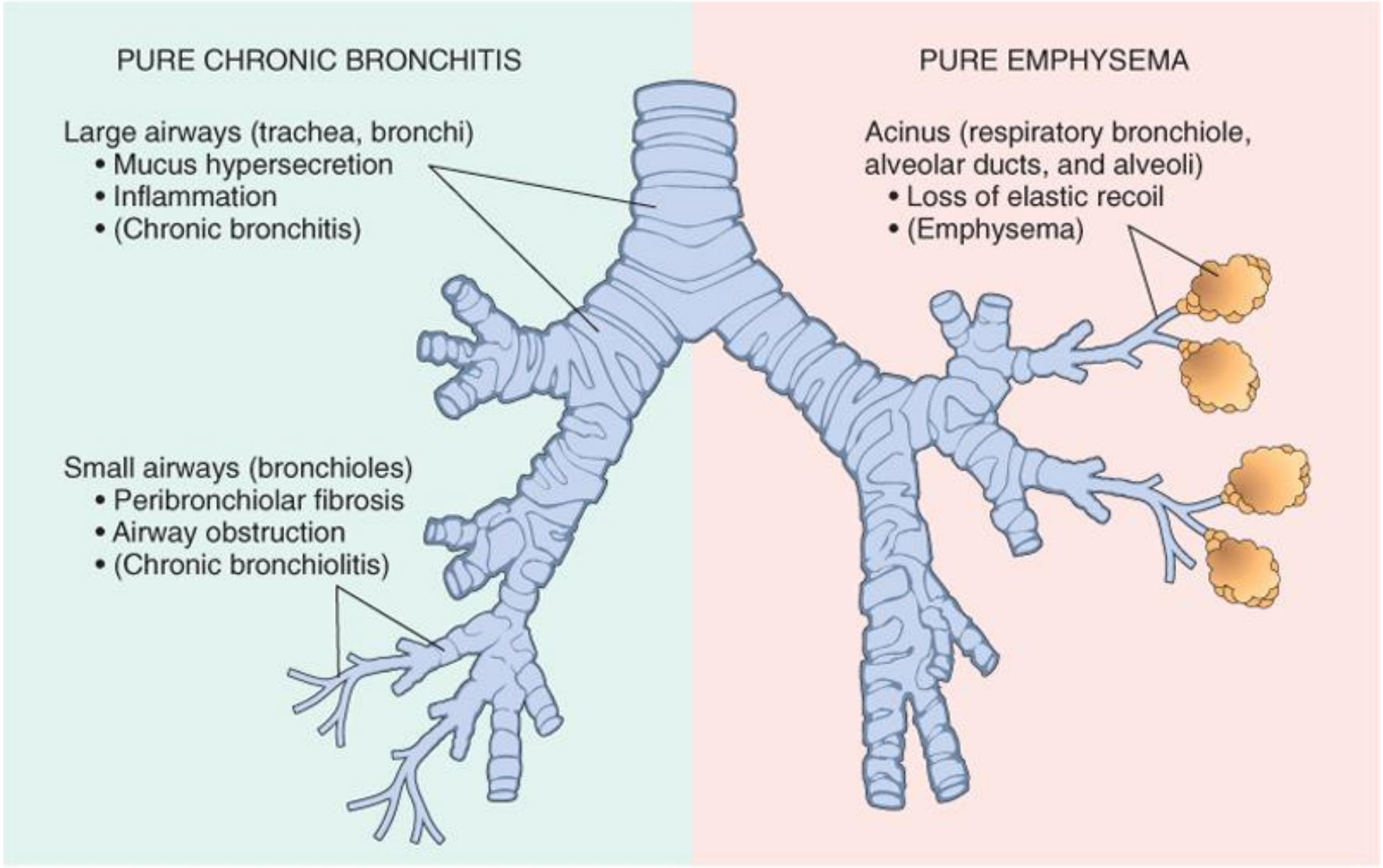
Morphology

- Enlargement of the mucus glands, increased number of goblet cells, loss of ciliated epithelial cells, squamous metaplasia, dysplastic changes and bronchogenic carcinoma.
- Inflammation, fibrosis and resultant narrowing of bronchioles.
- Coexistent emphysema

Clinical features

- Prominent cough and the production of sputum.
- COPD with hypercapnia,
- hypoxemia and cyanosis.
- Cardiac failure

Chronic bronchitis vs. Emphysema



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Emphysema and Chronic Bronchitis

	Predominant Bronchitis	Predominant Emphysema
Appearance	"Blue bloaters"	"Pink Puffers"
Age	40-45	50-75
Dyspnea	Mild, late	Severe, early
Cough	Early, copious sputum	Late, scanty sputum
Infection	Common	Occasional
Respiratory Insufficiency	Repeated	Terminal
Cor pulmonale	Common	Rare, terminal
Airway resistance	Increased	Normal or slightly increased
Elastic recoil	Normal	Low
Chest radiography	Prominent vessels, large heart	Hyperinflation, small heart

CHRONIC BRONCHITIS



BLUE BLOATER

Dyspnea on Exertion

Cough, DOE,
Hypercapnia,
Hypoxemia
Mild cyanosis

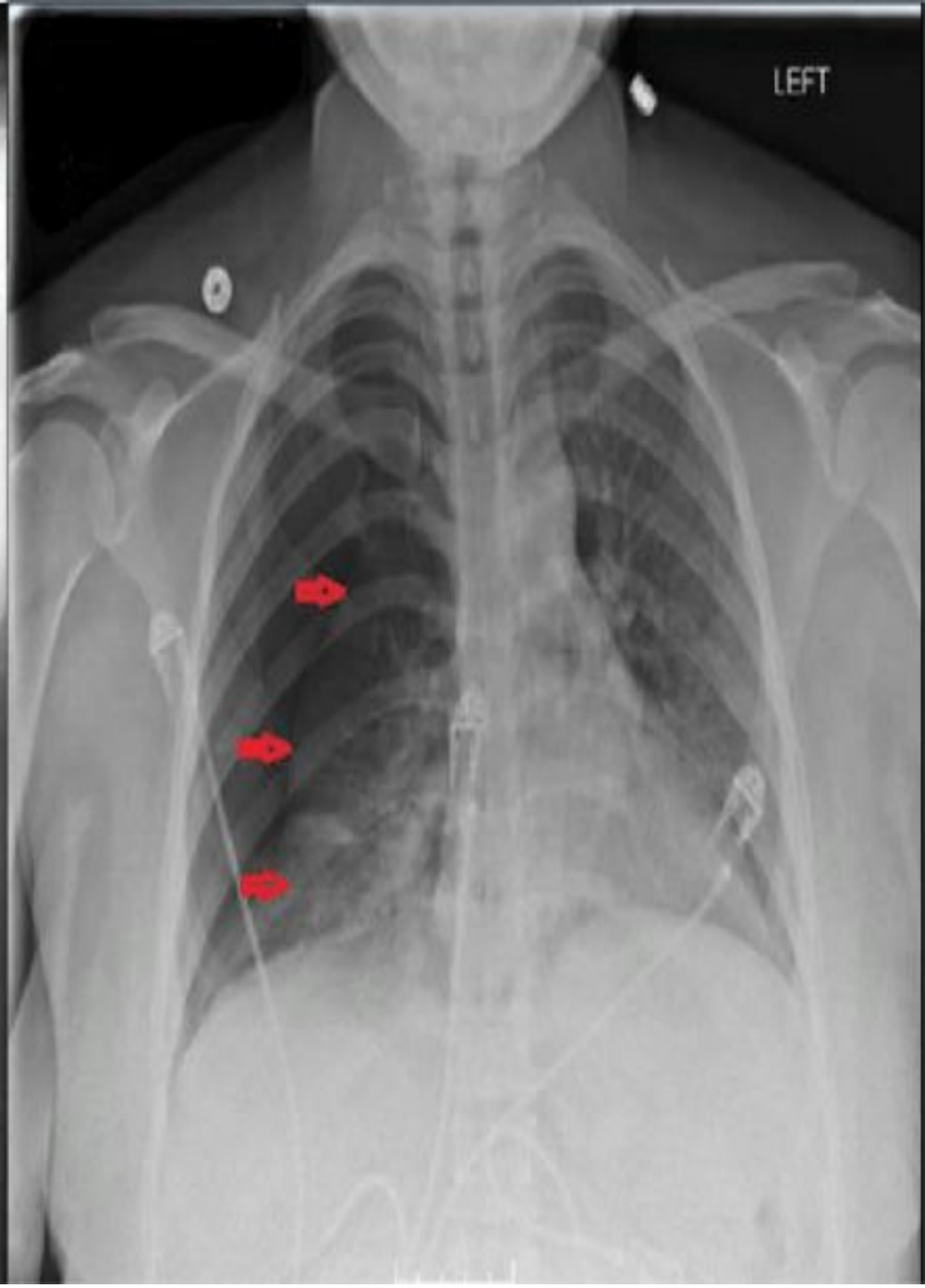
EMPHYSEMA



PINK PUFFER

Barrel-chested
Dyspneic
prolonged expiration,
Hunched-over position,
Breathes through pursed lips




LEFT

ASTHMA

Asthma: Dyspnea and wheezing

Types

1. Extrinsic asthma: Type 1 Hypersensitivity reaction, IgE, childhood, family Hx of allergy.
2. Intrinsic asthma: associated e bronchial asthma, aspirin, exercise, cold induced. No Hx of allergy

Morphology

- Hypertrophy of bronchial smooth muscle & hyperplasia of goblet cells eosinophils
- Mucous plug e Curschmann spirals & Charcot-Leyden crystals.

Complication

- Superimposed infection
- Chronic bronchitis
- Pulmonary emphysema
- Status asthmaticus

CLASSIFICATION OF ASTHMA

Extrinsic Asthma

- Initiated by type 1 hypersensitivity reaction induced by exposure to extrinsic antigen.
- Subtypes include:
  a. atopic (allergic) asthma.
  b. occupational asthma.
  c. allergic bronchopulmonary aspergillosis.
- Develop early in life

Intrinsic Asthma

- Initiated by diverse, non-immune mechanisms, including ingestion of aspirin, pulmonary infections, cold, inhaled irritant, stress and exercise.
- No personal or family history of allergic reaction.
- Develop later in life

Non-Atopic Asthma

- Triggered by respiratory tract infection including viruses and inhaled air pollutants e.g. sulfur dioxide, ozone.
- Positive family history is uncommon.
- Serum IgE – normal.
- No other associated allergies.
- Skin test – negative.
- Hyperirritability of bronchial tree.
- Subtypes:
  1. Drug-induced asthma.
  2. Occupational asthma.

Extrinsic Asthma

- Atopic (allergic) asthma is the most common form, begins in childhood
- Other allergic manifestation: allergic rhinitis, urticaria, eczema.
- Skin test with antigen result in an immediate wheel and flare reaction
- Other family member is also affected
- Serum IgE and eosinophil are increased
- immune related, T_H2 subset of CD4+ T cells

Clinical Course

- **Classic asthmatic attack** – dyspnea, cough, difficult expiration, progressive hyperinflation of lung and mucous plug in bronchi. This may resolve spontaneously or with Rx.
- **Status asthmaticus** – severe cyanosis and persistent dyspnea, may be fatal.
- May progress to emphysema.
- Superimposed bacterial infection may occur.

Bronchiectasis

Bronchiectasis

Chronic necrotizing infection of the bronchi and bronchioles leading to permanent dilation of these airways

Causes

- Infection
- Obstruction
- Congenital (Cystic fibrosis, Kartagener's Syndrome)

Clinical features

- Sever persistent cough with sputum (mucopurulent sputum) sometime with blood.
- Clubbing of fingers.

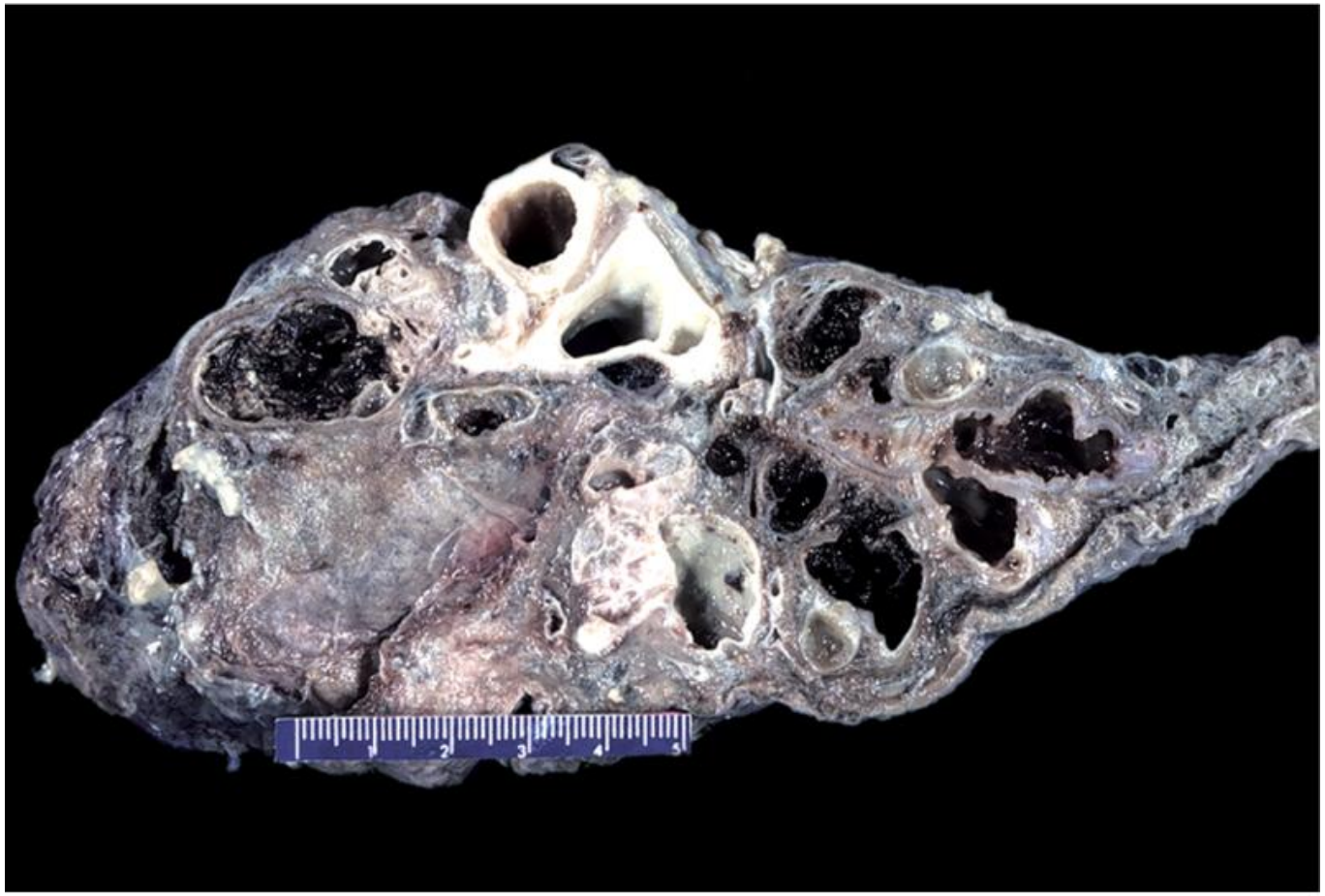
Complications

- If sever, obstructive pulmonary function develop.
- Lung Abscess
- Rare complications: metastatic brain abscess and amyloidosis.

Morphology of Bronchiectasis

- Usually affects lower lobes bilaterally (vertical airways).
- Dilated airways up to four times of normal, reaching the pleura.
- Tube-like enlargement (cylindroid) or fusiform (saccular).
- Acute and chronic inflammation, extensive ulceration of lining epithelium with fibrosis.

Bronchiectasis



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